

Modeling the 2004 Indian Ocean Tsunami Using BVPs and PDEs

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Table of Contents

- 1 Definition of Parameters and Variables
- 2 Results and Analysis
- 3 Error Analysis
- 4 Model Validation
- 5 Conclusion

Table of Contents

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- 2 Results and Analysis
- 3 Error Analysis
- 4 Model Validation
- 5 Conclusion

Boundary Value Problem

The BVP models the steady-state energy flux through the ocean: as seafloor rises toward shore and wave slows, conservation of energy forces wave displacement to increase.

$$\frac{d}{dx} \left[c(x)^2 \frac{dy}{dx} \right] = 0, \quad 0 \leq x \leq L$$

Boundary conditions:

$$y(0) = \alpha = 0, \quad y(L) = \beta = 1$$

$c(x)$ is defined as:

$$c(x) = \sqrt{g \cdot d(x)}, \quad d(x) = d_0 - \frac{d_0 - d_L}{L} x, \quad g = 9.81 \text{ m/s}^2$$

Shallow Water 1-D Wave Equation

The PDE models the propagation of the tsunami across the ocean, including the change in water depth across the x domain of the function.

$$\frac{\partial^2 \eta}{\partial t^2} = \frac{\partial}{\partial x} \left[c(x)^2 \frac{\partial \eta}{\partial x} \right], \quad 0 \leq x \leq L, \quad 0 \leq t \leq T$$

Boundary conditions:

$$\eta(0, t) = 0, \quad \eta(L, t) = 0, \quad t > 0$$

Initial conditions:

$$\eta(x, 0) = \begin{cases} A \left(1 - \frac{|x - x_0|}{w} \right) & |x - x_0| \leq w \\ 0 & |x - x_0| > w \end{cases}, \quad \frac{\partial \eta}{\partial t}(x, 0) = 0$$

Constants and Coefficients

Symbol	Description	Value	Units
L	Domain length (ocean transect)	5×10^5	m
T	Total simulation time	3,600	s
g	Gravitational acceleration	9.81	m/s ²
d_0	Water depth at $x = 0$	4,000	m
d_L	Water depth at $x = L$	10	m
α	BVP boundary condition at $x = 0$	0	m
β	BVP boundary condition at $x = L$	1	m
A	Initial wave amplitude	1	m
x_0	Initial wave center position	5×10^4	m
w	Initial wave half-width	2.5×10^4	m

Table of Contents

- 1 Definition of Parameters and Variables
- 2 Results and Analysis
- 3 Error Analysis
- 4 Model Validation
- 5 Conclusion

Numerical Method: Linear Shooting (BVP)

BVP is a linear 2nd-order ODE (Burden & Faires Algorithm 11.1).

Rewrite as two 1st-order IVPs with $y_1 = y$, $y_2 = y'$:

$$y_1' = y_2, \quad y_2' = -p(x) y_2, \quad p(x) = \frac{d_0 - d_L}{L \cdot d(x)}$$

Decompose into u -system (particular) and v -system (homogeneous):

$$u_1(0) = \alpha, \quad u_2(0) = 0 \quad \text{and} \quad v_1(0) = 0, \quad v_2(0) = 1$$

Solve systems with RK4. Assemble complete solution as:

$$y(x_i) = u_{1,i} + s \cdot v_{1,i}, \quad s = \frac{\beta - u_{1,N}}{v_{1,N}}$$

Result

$$s = 3.329826 \times 10^{-7}$$

Numerical Method: FTCS Finite Difference (PDE)

Modified Forward-Time, Centered-Space scheme (Burden & Faires Algorithm 12.2). Since $c(x)^2$ varies with x , wave speed is evaluated at half-nodes before the time loop:

$$c_{i+\frac{1}{2}}^2 = g \cdot d\left(\left(i + \frac{1}{2}\right) h\right), \quad c_{i-\frac{1}{2}}^2 = g \cdot d\left(\left(i - \frac{1}{2}\right) h\right)$$

New interior mesh update formula:

$$w_{i,j+1} = 2w_{i,j} - w_{i,j-1} + \left(\frac{k}{h}\right)^2 \left[c_{i+\frac{1}{2}}^2 (w_{i+1,j} - w_{i,j}) - c_{i-\frac{1}{2}}^2 (w_{i,j} - w_{i-1,j}) \right]$$

Grid parameters

$m = 500$ spatial intervals, $h = 1000$ m, $N = 900$ time steps, $k = 4$ s

BVP Results — $y(x)$

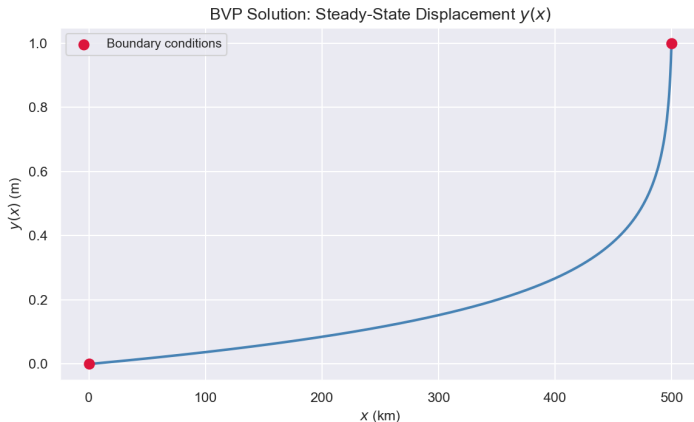


Figure: Plot showing numerical solution to $y(x) = u(x) + s \cdot v(x)$. Boundary conditions are shown in red. The plot shows a gradual increase initially that grows steeper over time.

BVP Results — $y'(x)$

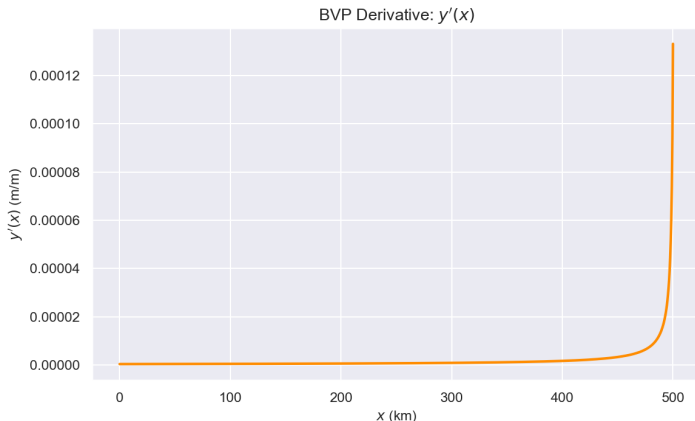


Figure: Plot showing numerical solution for $y'(x)$. Since $c(x)^2 y'(x) = \text{constant}$, as $d(x) \rightarrow 0$ near shore, $y'(x)$ increases sharply to maintain constant energy flux.

PDE Results — Wave Snapshots

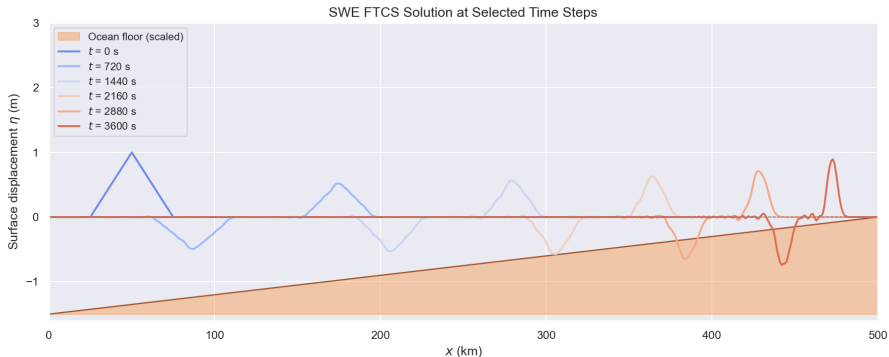


Figure: Snapshots of $\eta(x, t)$ over the time domain of the simulation. The wave begins as a wide triangular displacement and gradually evolves into a steeper, narrower wave as it approaches shore.

PDE Results — Final State at $t = T$

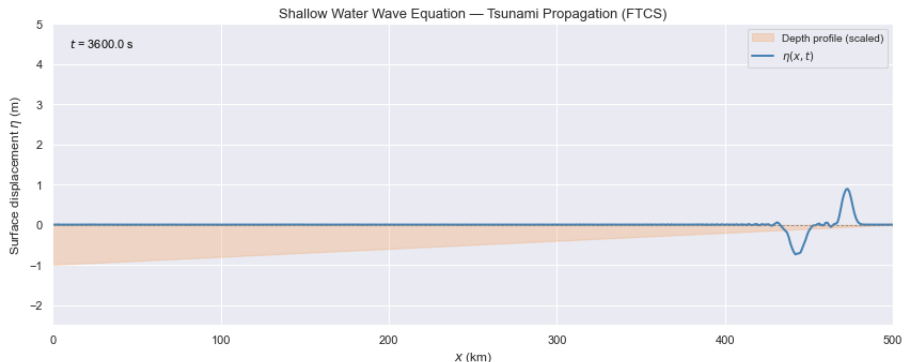


Figure: Solution $\eta(x, T)$ at the end of the simulation ($t = 3600$ s), showing the wave concentrated and amplified in the shallowest region near the shoreline.

Step-Size Sensitivity Analysis

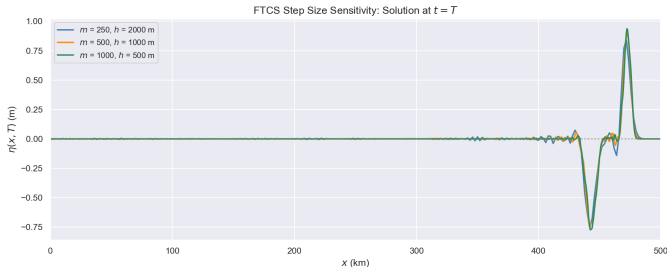


Figure: Overlapping FTCS solutions at $t = T$ for three spatial resolutions ($m = 250, 500, 1000$).

Comparison	RMSE (m)
$m = 250$ vs $m = 500$	2.629×10^{-2}
$m = 500$ vs $m = 1000$	1.312×10^{-2}
Convergence ratio r	2.004

Max difference between resolutions is $< 2.6\%$ of wave amplitude. Shows solution is numerically stable among range of step sizes.

Table of Contents

- 1 Definition of Parameters and Variables
- 2 Results and Analysis
- 3 Error Analysis**
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- 5 Conclusion

BVP Error Analysis — Analytical Solution

Since $c(x)^2 y'(x) = K$ (constant flux), integrating and applying boundary conditions yields:

$$y(x) = \frac{\ln(d(x)/d_0)}{\ln(d_L/d_0)} \approx \frac{\ln(d(x)/4000)}{-5.9915}$$

Numerical and analytical solutions are nearly identical. Absolute error peaks near $x = L$ where the solution changes most rapidly.

Root Mean Squared Error

$$\text{RMSE} = 6.034 \times 10^{-6} \text{ m}$$

Error is on the order of 10^{-6}m across 500 km domain.

Convergence ratio

$$r = \frac{\text{RMSE}(m_{\text{coarse}}, m_{\text{medium}})}{\text{RMSE}(m_{\text{medium}}, m_{\text{fine}})} = \frac{2.629 \times 10^{-2}}{1.312 \times 10^{-2}} \approx 2.00$$

Interpretation: Shrinking step size by factor of 2 corresponds with halving error = first-order convergence in h .

Standard FTCS with constant is second-order convergent ($r \approx 4$).

Reduction to first-order is due to truncation error at each node due to the half-node interpolation of $c(x)^2$.

Table of Contents

- 1 Definition of Parameters and Variables
- 2 Results and Analysis
- 3 Error Analysis
- 4 Model Validation**
- 5 Conclusion

BVP Validation

- Analytical solution derived independently
- Numerical RMSE = 6.03×10^{-6} m against exact solution
- Boundary conditions exactly satisfied by linear shooting solution: $y(0) = 0$, $y(L) = 1$

PDE Validation

- Three resolutions produce $< 2.6\%$ difference at $t = T$
- Convergence ratio $r \approx 2$ shows systematic, explainable error reduction

Table of Contents

- 1 Definition of Parameters and Variables
- 2 Results and Analysis
- 3 Error Analysis
- 4 Model Validation
- 5 Conclusion**

Conclusion

Model Interpretation

As a tsunami enters shallow water, conservation of energy forces wave amplitude to increase sharply. The destructive power of the 2004 Indian Ocean Tsunami can be at least partially explained by the increase in vertical wave displacement due to the shallower water depth.

Takeaways

- Linear shooting method with RK4 is highly accurate ($\text{RMSE} \sim 10^{-6}$ m) and required no algorithmic modification
- FTCS scheme required adaptation for variable $c(x)$ (half-nodes)
- FTCS achieves only first-order convergence due to variable $c(x)$

References



R. L. Burden and J. D. Faires, *Numerical Analysis*, 9th ed. Brooks/Cole, Cengage Learning, 2011.



K. Murase, “Wave-Propagation Velocity, Tsunami Speed, Amplitudes,” in *Proc. 15th World Conference on Earthquake Engineering (15WCEE)*, Lisbon, Portugal, 2012, Paper No. 0335. Available: https://www.iitk.ac.in/nicee/wcee/article/WCEE2012_0335.pdf